Test Series No.

1.6

MASTERY TEST SERIES

DURATION: 90 Minutes | **M. MARKS: 400**

DATE: 24/02/2026

ANSWER KEY

1. (3)	26. (1)	51. (3)	76. (4)
2. (2)	27. (3)	52. (2)	77. (3)
3. (2)	28. (2)	53. (2)	78. (4)
4. (1)	29. (3)	54. (4)	79. (3)
5. (3)	30. (2)	55. (2)	80. (3)
6. (1)	31. (4)	56. (1)	81. (4)
7. (1)	32. (1)	57. (2)	82. (3)
8. (1)	33. (2)	58. (2)	83. (4)
9. (3)	34. (3)	59. (4)	84. (3)
10. (3)	35. (4)	60. (4)	85. (3)
11. (2)	36. (1)	61. (4)	86. (1)
12. (3)	37. (3)	62. (1)	87. (1)
13. (2)	38. (3)	63. (2)	88. (2)
14. (4)	39. (3)	64. (2)	89. (4)
15. (3)	40. (2)	65. (4)	90. (3)
16. (3)	41. (1)	66. (2)	91. (3)
17. (3)	42. (2)	67. (1)	92. (3)
18. (4)	43. (1)	68. (1)	93. (4)
19. (4)	44. (4)	69. (3)	94. (3)
20. (3)	45. (3)	70. (4)	95. (4)
21. (2)	46. (4)	71. (2)	96. (4)
22. (4)	47. (1)	72. (4)	97. (4)
23. (1)	48. (4)	73. (2)	98. (3)
24. (3)	49. (1)	74. (2)	99. (4)
25. (3)	50. (1)	75. (2)	100. (4)

[Contd...

Hints & Solutions

1. (3)

In fusion reaction, light nuclei first come close to each other after that they will fuse. They overcome coloumbic repulsion due to high temperature.

(NCERT Ed. 2025-26, 12th, Page No. 316)

2. (2)

$$Y = \overline{\overline{A \cdot B}} + \overline{B}$$

$$= A \cdot B + \overline{B}$$

3. (2)

According to Gauss's law:

$$\phi_{\text{Total}} = \frac{q_{\text{enclosed}}}{\varepsilon_0}$$

As charge on α-particle = $2e$

$$\therefore \ \phi_T = \frac{2e}{\varepsilon_0}$$

(NCERT Ed. 2025-26, 12th, Page No. 30)

4. (1)

Number of pulse per second in output is same as number of pulses per second in input.

(NCERT Ed. 2025-26, 12th, Page No. 338)

5. (3)

Path difference corresponds to dark fringe is given by

$$\Delta x = (2n-1)\frac{\lambda}{2},\ n = 1, 2, 3....$$

For 5th dark fringe

$$\Delta x = \frac{9\lambda}{2}$$

$$\Delta\phi = \frac{2\pi}{\lambda}\Delta x = \frac{2\pi}{\lambda} \times \frac{9\lambda}{2} = 9\pi$$

(NCERT Ed. 2025-26, 12th, Page No. 266)

6. (1)

Path difference at centre of screen is always zero. Hence, all wavelengths always interfere constructively at centre.

(NCERT Ed. 2025-26, 12th, Page No. 265)

7. (1)

$$\phi = B.A$$

$$= at.\pi r^2$$

$$\varepsilon = \frac{-d\phi}{dt} = -\pi r^2 a$$

(NCERT Ed. 2025-26, 12th, Page No. 158)

8. (1)

$$V = 2.18 \times 10^6 \times \frac{Z}{n}$$

$$= \frac{2.18 \times 10^6 \times 2}{3}$$

$$= 1.46 \times 10^6 \text{ m/s}$$

(NCERT Ed. 2025-26, 12th, Page No. 296)

9. (3)

Net magnetic dipole moment $= 2M\cos\frac{\theta}{2}$

As value of $\cos\frac{\theta}{2}$ is maximum in case (3) hence net magnetic dipole moment is maximum for option (3).

(NCERT Ed. 2025-26, 12th, Page No. 151)

10. (3)

From Malu's law, the intensity of light transmitted through an analyser is given by

$$I = I_0 \cos^2\theta$$

$$\frac{I}{I_0} = \cos^2 60° \ \Rightarrow \ I = \frac{I_0}{4}$$

(NCERT Ed. 2025-26, 12th, Page No. 270)

11. (2)

Irrespective of the photo sensitive material used, the slope of each line should be same.

Since, $W_A > W_B > W_C$; $(v_0)_A > (v_0)_B > (v_0)_C$

Thus, the correct order is *C*, *B*, *A*.

Option (3) and (4) are not possible since the graphs are not parallel.

(NCERT Ed. 2025-26, 12th, Page No. 279)

12. (3)

$$\lambda = \frac{h}{\sqrt{2mK}};$$

$$\therefore \ \lambda \propto \frac{1}{\sqrt{K}} \ \therefore \ \frac{\lambda'}{\lambda} = \sqrt{\frac{K}{4K}} = \frac{1}{2} \text{ or } \lambda' = \frac{\lambda}{2}$$

(NCERT Ed. 2025-26, 12th, Page No. 285)

[Contd...

13. (2)

The hydrogen nucleus consists of only one proton.

Therefore, its binding energy per nucleon is zero.

(NCERT Ed. 2025-26, 12th, Page No. 312)

14. (4)

$$v = \frac{c}{\sqrt{\mu_r \varepsilon_r}} = \frac{3\times10^8}{\sqrt{8\times2}} = \frac{3}{4}\times10^8$$

$$v = 7.5\times10^7 \text{ m/s}$$

(NCERT Ed. 2025-26, 12th, Page No. 207)

15. (3)

Resonance condition is not possible in series *RL* or *RC* circuit. Transformer is based on principle of mutual induction.

$N_S > N_P$ for step-up transformer.

(NCERT Ed. 2025-26, 12th, Page No. 191)

16. (3)

Both 6 Ω resistors are shorted.

$$i_{PQ} = \frac{20}{5} = 4\text{ A}$$

(NCERT Ed. 2025-26, 12th, Page No. 101)

17. (3)

$$\frac{(E_K)_1}{(K_K)_2} = \frac{h\nu_1 - \phi_0}{h\nu_2 - \phi_0} = \frac{1-0.5}{2.5-0.5} = \frac{0.5}{2} = \frac{1}{4}$$

(NCERT Ed. 2025-26, 12th, Page No. 281)

18. (4)

$${}_1^2H + {}_1^2H \rightarrow {}_2^4He + E$$

$$(B.E)_{\text{reactant}} = 2(1.1+1.1) = 4.4\text{ MeV}$$

$$(B.E)_{\text{product}} = 4\times7 = 28\text{ MeV}$$

$E = 28 - 4.4 = 23.6$ MeV

(NCERT Ed. 2025-26, 12th, Page No. 315)

19. (4)

For photo-diode and zener-diode, *I-V* characteristic lies in III quadrant and for solar cell, *I-V* characteristic lies in IV quadrant.

20. (3)

$$\frac{[h]}{[I]} = \frac{\left[\frac{ML^2T^{-2}}{T^{-1}}\right]}{[ML^2]} = [T^{-1}] = [\text{frequency}]$$

(NCERT Ed. 2025-26, 11th, Page No. 7)

21. (2)

$$W_{WXY} = W_{WX} + W_{XY}$$

$$= 0 + (100\times10^3\times300\times10^{-6}) = 30 \text{ Joule}$$

(NCERT Ed. 2025-26, 11th, Page No. 231)

22. (4)

$$L = \frac{nh}{2\pi}$$

$$\frac{L_1}{L_2} = \frac{3}{5}$$

(NCERT Ed. 2025-26, 12th, Page No. 299)

23. (1)

Angular width of the central bright is $\frac{2\lambda}{d}$

$$= \frac{2\times6328\times10^{-10}}{0.2\times10^{-3}}\times\frac{180^\circ}{\pi} = 0.36 \text{ degree}$$

(NCERT Ed. 2025-26, 12th, Page No. 267)

24. (3)

$$\frac{1}{\lambda} = R\left(\frac{1}{n_f^2} - \frac{1}{n_i^2}\right)$$

For longest wavelength, $n_i = 2,\ n_f = 1$

$$\frac{1}{\lambda_1} = R\left(\frac{1}{1^2} - \frac{1}{2^2}\right) = \frac{3R}{4} \quad \text{.....(i)}$$

For shortest wavelength, $n_i = \infty,\ n_f = 1$

$$\frac{1}{\lambda_2} = R\left(\frac{1}{1^2} - \frac{1}{\infty^2}\right) = R \quad \text{.....(ii)}$$

$$\frac{\lambda_1}{\lambda_2} = \frac{R}{\frac{3R}{4}} = \frac{4}{3}$$

(NCERT Ed. 2025-26, 12th, Page No. 301)

25. (3)

Lorentz force $= q(\vec{v}\times\vec{B})$

$$-8\hat{k} = q\begin{vmatrix}\hat{i} & \hat{j} & \hat{k}\\ 2 & 3 & 0\\ 0 & 2 & 0\end{vmatrix}\times10^6$$

$$-8\hat{k} = q\{\hat{i}(0) - \hat{j}(0) + \hat{k}(4-0)\}\times10^6$$

$$-8\hat{k} = 4q\hat{k}\times10^6$$

$$\therefore 4q = -8\times10^{-6}$$

$q = -2\times10^{-6}C$

$= -2\mu C$

(NCERT Ed. 2025-26, 12th, Page No. 109)

[Contd...

26. (1)

The formula to calculate spin only magnetic moment is:

$\sqrt{n(n+2)}$

Here, n = no. of unpaired electrons

$\sqrt{1(1+2)} = 1.73$

$V^{2+} = 3\ d^3$

$\sqrt{3(3+2)} = 3.87$

$Ni^{2+} = 3\ d^8 = 2$ unpaired electrons

$\sqrt{2(2+2)} = 2.84$

$Sc^{3+} = 3\ d^0$

$Sc^{3+} = 0$

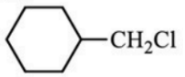
27. (3)

A catalyst catalyses only spontaneous reactions.

The term $e^{-\frac{E_a}{RT}}$ represents the fraction of molecules with energies equal or greater than E_a.

(NEW NCERT 12th Part-I Page No. 82, 83)

28. (2)



is 1° halide and the lone pair of Cl is not involve in resonance so undergoes S_N2 reaction faster.

(NEW NCERT 12th Part-II Page No. 174)

29. (3)

The only well characterised compound of Bi(V) is BiF_5

30. (2)

More the CFSE, lesser will be the wavelength absorb.

Strong field ligand will show more CFSE.

Ligand field strength: $NH_3 > H_2O > Cl^-$

(NEW NCERT 12th Part-II Page No. 132)

31. (4)

There are no significant uses of Xenon and Krypton. They are used in light bulbs designed for special purposes.

Helium is used in filling balloons for metrological observations.

32. (1)

With an increase in oxygen percentage, an element's oxide becomes more acidic and less basic.

Hence, the correct order is given by option (1).

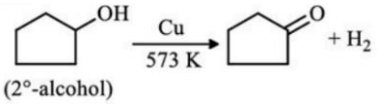
33. (2)

$[(Co)_2(CO)_8]$ is an organometallic compound as it contain metal-carbon bond.

Metallic carbonyl involve synergic bonding.

(NEW NCERT 12th Part-II Page No. 136)

34. (3)



(2°-alcohol)

(NEW NCERT 12th Part-II Page No. 210)

35. (4)

The **incorrect** statements are:-

(D) Due to inert pair effect, lower oxidation state is more stable.

(E) Nitrogen belongs to 2nd period and cannot expands its octet.

36. (1)

M_O = [Kr]$4d^5 5s'$, *d* electrons = 5 = x

Rh = [Kr]$4d^8 5'$, *d* electrons = 8 = y

x + y = 5 + 8 = 13

37. (3)

Among group-16 hydrides acidic character increases from H_2O to H_2Te because bond energy decreases on moving down the group.

38. (3)

Complex in which a metal is bound to more than one kind of donor groups, eg $[Co(NH_3)_4Cl_2]$ are known as heteroleptic.

$[Cu(CN)_4]^{3-}$: x + 4(-1) = -3

x = +1

(NEW NCERT 12th Part-I Page No. 122)

39. (3)

- Thermal decomposition of HI on gold surface is an example of zero order reaction.

[Contd...

- Natural radioactive decay follows first order kinetics.
- Unit of rate constant for second order reaction is
  $M^{-1} s^{-1}$.
- Inversion of cane sugar is an example of pseudo first order reaction.

(NEW NCERT 12th Part-I Page No. 72, 73, 78, 69)

40. (2)

Aryl halides are less reactive towards nucleophilic substitution reaction than alkyl halides because in aryl halides, C-Cl bond acquires a partial double bond character.

(NCERT Ed. 2025-26 11th Part-II Page No. 181)

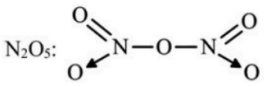
41. (1)

(A) Zn, Cd, Hg exhibit lowest enthalpy of atomisation in respective transition series.

(C) Compounds of Zn, Cd and Hg are diamagnetic in nature.

42. (2)

N_2O_5:



43. (1)

$[Ma_3b_3]$ complex exist as facial and meridional isomers.

(NEW NCERT 12th Part-I Page No. 126)

44. (4)

(Cations)	Group No.	Group reagent
Pb^{2+}, Cu^{2+}	I	H_2S gas in presence of dilute HCl
Al^{3+}, Fe^{3+}	III	NH_4OH in presence of NH_4Cl
Co^{2+}, Ni^{2+}	IV	H_2S in presence of NH_4OH
Ba^{2+}, Ca^{2+}	II	$(NH_4)_2CO_3$ in presence of NH_4OH

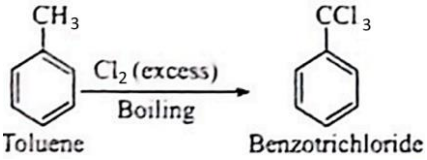
45. (3)

Melting point:

$H_2O > H_2Te > H_2Se > H_2S$

46. (4)

Side chain reaction occurs forming benzotrichloride.

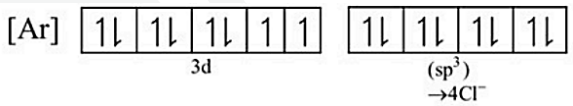


47. (1)

The orange colour of $K_2Cr_2O_7$ and purple colour of $KMnO_4$ is due to charge transfer transitions.

48. (4)

Ni^{2+} in $[NiCl_4]^{2-}$:



(NEW NCERT 12th Part-I Page No. 129)

49. (1)

$$CH_3COO^- + FeCl_3 \rightarrow Fe(CH_3COO)_3$$

or $[Fe_3(OH)_2(CH_3COO)_6]^+$

Blood red colour

$\downarrow \Delta$

$Fe(OH)_2(CH_3COO)\downarrow$

Red-brown precipitate

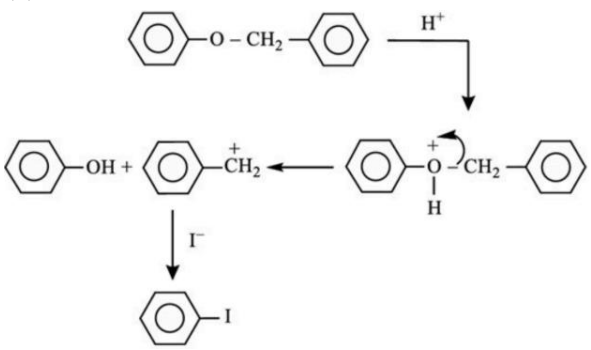
Brown ring test:

$$2NO_3^- + 4H_2SO_4 + 6Fe^{2+} \rightarrow 6Fe^{3+} + 2NO\uparrow + 4SO_4^{2-} + 4H_2O$$

$$\left[Fe(H_2O_6)\right]^{2+} + NO \rightarrow \left[Fe(H_2O)_5(NO)\right]^{2+} + H_2O$$

Brown

50. (1)



(NEW NCERT 12th Part-II Page No. 217)

[Contd...

51. (3)

Centrosomes undergo duplication during the S phase, but this occurs specifically in animal cells. Higher plant cells typically lack centrosomes/centrioles. Centrioles (and the centrosomes) begin to move towards opposite poles during prophase, not metaphase. The centrosome is a non-membrane bound organelle consisting of two cylindrical structures called centrioles.

(NCERT Ed. 2025-26, 11th, Page No. 122)

52. (2)

Forest, grassland and desert are some examples of terrestrial ecosystems; pond, lake, wetland, river and estuary are some examples of aquatic ecosystems. Crop fields and an aquarium may also be considered as man-made ecosystems.

(NCERT Ed. 2025-26, 12th, Page No. 206)

53. (2)

A population growing in a habitat with limited resources show initially a lag phase, followed by phases of acceleration and deceleration and finally an asymptote, when the population density reaches the carrying capacity. A plot of N in relation to time (t) results in a sigmoid curve. This type of population growth is called Verhulst-Pearl Logistic Growth and is described by the following equation:

$$\frac{dN}{dt} = rN\frac{(K-N)}{K}$$

(NCERT Ed. 2025-26, 12th, Page No. 195)

54. (4)

The earliest systems of classification used only gross superficial morphological characters such as habit, colour, number and shape of leaves, etc. They were based mainly on vegetative characters or on the androecium structure (system given by Linnaeus). Such systems were artificial.

(NCERT Ed. 2025-26, 11th, Page No. 23)

55. (2)

A single species might show high diversity at the genetic level over its distributional range. The genetic variation shown by the medicinal plant *Rauwolfia vomitoria* growing in different Himalayan ranges might be in terms of the potency and concentration of the active chemical (reserpine) that the plant produces.

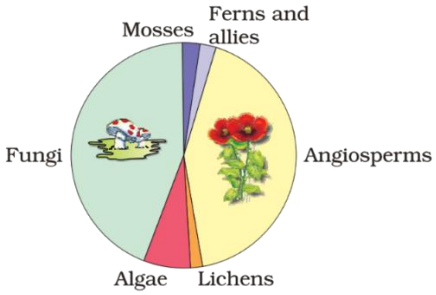
(NCERT Ed. 2025-26, 12th, Page No. 217)

56. (1)

RuBisCO	C_3 cycle
PEPcase	C_4 cycle
Hexokinase	Glycolysis
Citrate synthase	Krebs cycle

(NCERT Ed. 2025-26, 11th, Page No. 143)

57. (2)



A pie chart representing the proportionate number of species of major plant taxa.

(NCERT Ed. 2025-26, 12th, Page No. 218)

58. (2)

The human liver fluke (a trematode parasite) depends on two intermediate hosts (a snail and a fish) to complete its life cycle.

(NCERT Ed. 2025-26, 12th, Page No. 200)

59. (4)

In dicot stems, the hypodermis consists of collenchyma cells while in monocot stems, the hypodermis is made of sclerenchyma cells. Dicot leaves generally exhibit reticulate venation, whereas monocot leaves usually show parallel venation. Most dicot seeds are non-endospermous at maturity (e.g., pea, gram, bean), while most monocot seeds (e.g., maize, wheat) are endospermous. In all angiosperm roots (both monocot and dicot), the protoxylem (the first formed primary xylem) lies towards the periphery (outside), while the metaxylem (later formed) lies towards the centre (pith). This specific orientation is termed exarch.

(NCERT Ed. 2025-26, 11th, Page No. 74)

60. (4)

According to the 10% Law of energy transfer, only 10% of the energy from one trophic level is transferred to the next.

Energy transfer in an ecosystem follows a specific sequence of trophic levels:

Producers (T_1): 4000 J (given)

Primary consumers / Herbivores (T_2): 10% of 4000 J = 400 J

Secondary consumers / Primary carnivores (T_3): 10% of 400 J = 40 J

(NCERT Ed. 2025-26, 12th, Page No. 211)

61. (4)

Insertion or deletion of one or two bases changes the reading frame from the point of insertion or deletion. However, inserting or deleting three bases (or multiples of three) does not change the reading frame; instead, it adds or deletes one or more amino acids without shifting the subsequent reading frame.

(NCERT Ed. 2025-26, 12th, Page No. 98)

[Contd...

62. (1)
Although all the biodiversity hotspots put together cover less than 2 per cent of the earth's land area, the number of species they collectively harbour is extremely high and strict protection of these hotspots could reduce the ongoing mass extinctions by almost 30 per cent.
(NCERT Ed. 2025-26, 12th, Page No. 225)

63. (2)

Species A	Species B	Name of Interaction
+	+	Mutualism
-	-	Competition
+	-	Predation
+	-	Parasitism
+	0	Commensalism
-	0	Amensalism

(NCERT Ed. 2025-26, 12th, Page No. 197)

64. (2)
Cleistogamous flowers which do not open at all. In such flowers, the anthers and stigma lie close to each other. When anthers dehisce in the flower buds, pollen grains come in contact with the stigma to effect pollination. Thus, cleistogamous flowers are invariably autogamous as there is no chance of cross-pollen landing on the stigma. Cleistogamous flowers produce assured seed-set even in the absence of pollinators.
(NCERT Ed. 2025-26, 12th, Page No. 12)

65. (4)
The humus is further degraded by some microbes and release of inorganic nutrients occur by the process known as mineralisation.
(NCERT Ed. 2025-26, 12th, Page No. 208)

66. (2)
Statins produced by the yeast *Monascus purpureus* have been commercialised as blood-cholesterol lowering agents.
(NCERT Ed. 2025-26, 12th, Page No. 153)

67. (1)

Colombia	Nearly 1,400
New York	105
Greenland	56
India	More than 1,200

(NCERT Ed. 2025-26, 12th, Page No. 219)

68. (1)
Predation is nature's way of transferring to higher trophic levels the energy fixed by plants. The intrinsic rate of natural increase is a measure of the inherent potential of a population to grow.
(NCERT Ed. 2025-26, 12th, Page No. 197)

69. (3)
Cytokinins were discovered as kinetin (a modified form of adenine, a purine) from the autoclaved herring sperm DNA.
(NCERT Ed. 2025-26, 11th, Page No. 176)

70. (4)
The pyramid of biomass in sea is generally inverted because the biomass of fishes far exceeds that of phytoplankton.
The pyramid of energy is always upright.
In a grassland ecosystem, the pyramid of numbers is upright (many grasses support fewer herbivores, which support even fewer carnivores).
(NCERT Ed. 2025-26, 12th, Page No. 213)

71. (2)
India, with about 45,000 species of plants and twice as many species of animals, is one of the 12 mega diversity countries of the world.
(NCERT Ed. 2025-26, 12th, Page No. 219)

72. (4)
Plants have evolved an astonishing variety of morphological and chemical defences against herbivores. Thorns (*Acacia, Cactus*) are the most common morphological means of defence. Many plants produce and store chemicals that make the herbivore sick when they are eaten, inhibit feeding or digestion, disrupt its reproduction or even kill it. You must have seen the weed *Calotropis* growing in abandoned fields. The plant produces highly poisonous cardiac glycosides and that is why you never see any cattle or goats browsing on this plant. A wide variety of chemical substances that we extract from plants on a commercial scale (nicotine, caffeine, quinine, strychnine, opium, etc.,) are produced by them actually as defences against grazers and browsers. Showing sexual deceit is a mechanism used for pollination, not a defense mechanism against herbivores.
(NCERT Ed. 2025-26, 12th, Page No. 198)

73. (2)
More than 70 per cent of all the species recorded are animals, while plants (including algae, fungi, bryophytes, gymnosperms and angiosperms) comprise no more than 22 per cent of the total. Among animals, insects are the most species-rich taxonomic group, making up more than 70 per cent of the total. That means, out of every 10 animals on this planet, 7 are insects.
(NCERT Ed. 2025-26, 12th, Page No. 218)

74. (2)
In an aquatic ecosystem, GFC is the major conduit for energy flow. As against this, in a terrestrial ecosystem, a much larger fraction of energy flows through the detritus food chain than through the GFC.
(NCERT Ed. 2025-26, 12th, Page No. 210)

75. (2)
The Nile perch introduced into Lake Victoria in east Africa led eventually to the extinction of an ecologically unique assemblage of more than 200 species of cichlid fish in the lake.
In the context of India, *Parthenium, Eichhornia, Lantana,* and *Clarias gariepinus* are all recognized as invasive alien species that have caused environmental damage.
(NCERT Ed. 2025-26, 12th, Page No. 223)

[Contd...

76. (4)

Hind II was the first restriction endonuclease characterized but it recognizes a specific sequence of six base pairs, not five. All other statements are correct—restriction enzymes are nucleases, exonucleases remove nucleotides from DNA ends, and endonucleases make cuts within DNA at specific sites.

(NCERT Ed. 2025-26, 12th, Page No. 165)

77. (3)

"Bt" in Bt cotton stands for *Bacillus thuringiensis*, a soil bacterium. This bacterium produces a protein toxin called Bt-toxin (Cry protein) that is lethal to insect larvae, particularly lepidopterans like cotton bollworms. The gene encoding this toxin is introduced into cotton plants to make them resistant to pest damage.

(NCERT Ed. 2025-26, 12th, Page No. 179)

78. (4)

The Pvu II restriction site is present in the rop gene of the pBR322 cloning vector. The pBR322 vector contains important features: the ori (origin of replication) controls DNA replication, amp^R provides ampicillin resistance, tet^R provides tetracycline resistance, and rop encodes a repressor protein. Pvu II cuts within the rop gene, allowing for cloning of foreign DNA at this site.

(NCERT Ed. 2025-26, 12th, Page No. 169)

79. (3)

Recombinant insulin is the biotechnological product that requires clinical trials before human use. Unlike Bt cotton (an agricultural product), Golden rice (a food crop), and Flavr Savr tomato (a food product), recombinant insulin is a therapeutic pharmaceutical drug. All drugs must undergo rigorous clinical trials to ensure safety and efficacy before approval for human administration.

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80. (3)

ADA deficiency can be permanently cured through gene therapy in early embryonic stages because the inserted gene gets integrated into germ cells and is inherited by all descendant cells. However, Reason R is false because lymphocytes obtained from the patient's blood are not immortal; they are normal cells with limited lifespan and require periodic infusion in gene therapy, not permanent cure.

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81. (4)

"Rosie," the first transgenic cow produced in 1997, was significant because her milk contained human alpha-lactalbumin. This was a breakthrough in pharmaceutical protein production through milk (a process called "pharming"). Alpha-lactalbumin is normally found in all mammalian milk, but the transgenic cow's milk contained the human form, making it valuable for infant nutrition and medical applications.

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82. (3)

List-I	List-II
Cotton bollworms	*cry*IAc and *cry*IIAb
Corn borer	*cry*IAb
Tobacco roots	*Meloidegyne incognitia*
Lepidopterans	Tobacco budworm

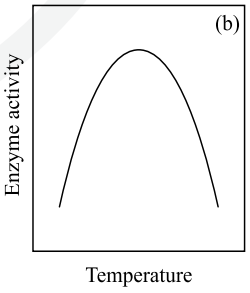
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83. (4)

Condom is a disposable barrier method, Vaults are a reusable barrier method, and LNG-20 is a hormonal contraceptive method. A condom is used once and discarded, vaults can be reused multiple times, and LNG-20 (levonorgestrel-releasing intrauterine device) is a hormonal contraceptive that provides long-term protection by releasing hormones directly into the reproductive tract.

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84. (3)



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85. (3)

Regarding *Eco*RI naming, Roman numerals indicate the order in which restriction enzymes were isolated from a specific bacterial strain. *Eco*RI comes from *Escherichia coli* (E = genus initial, co = species abbreviation, R = strain designation, and Roman numeral I = the first enzyme isolated from strain R). This standardized nomenclature helps scientists identify and classify different restriction enzymes systematically.

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86. (1)

Plasmodium, a tiny protozoan parasite causes malaria in humans, and is transmitted through the bite infected female *Anopheles* mosquito. When female *Anopheles* sucks the blood of infected human it takes up gametocytes (sexual stages of parasite) with blood meal. The gametocytes come out of the RBCs into the lumen cavity) of the stomach of the mosquito. In the stomach, the male gametocyte divides and forms 6 to 8 long, motile, whip-like microgametes (male gametes). The female gametocyte does not divide but undergoes a process maturation to become the macrogamete (female gamete). A microgamete penetrates a macrogamete and fertilisation (syngamy) takes place, resulting in the formation of a zygote. The zygote elongates and becomes worm like motile organism called ookinete. Ookinete further changes into sporozoites (mature infective stage *Plasmodium*).

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87. (1)

List-I (Technique)	List-II (Description)
Electrophoresis	Separation of DNA fragments
Elution	Extracting DNA from agarose gel
Transformation	Introducing DNA into a host
PCR	*In vitro* DNA amplification

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88. (2)

A probe in molecular diagnosis is a single-stranded DNA or RNA molecule tagged with a radioactive molecule (or fluorescent marker). This probe is complementary to a target DNA or RNA sequence and binds to it through base pairing, allowing scientists to detect and locate specific genes or mutations. The radioactive tag makes the bound probe visible during detection procedures.

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89. (4)

Intracytoplasmic sperm injection (ICSI) helps in both male infertility due to low sperm count and female infertility due to blocked fallopian tubes. In ICSI, a single healthy sperm is injected directly into an egg's cytoplasm, fertilizing it. The resulting embryo is then transferred to the uterus, bypassing the need for fallopian tubes and requiring only a single viable sperm, making it ideal for severe male factor infertility.

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90. (3)

DNA cannot pass through cell membranes easily because DNA is a large, hydrophilic (water-loving) molecule due to its phosphate backbone carrying negative charges, not because it is hydrophobic. The negative charges make DNA highly polar and unable to cross the lipid bilayer of the membrane, requiring special methods like electroporation or microinjection for cell transformation.

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91. (3)

PCR (Polymerase Chain Reaction) is the technique used to detect mutations in genes of suspected cancer patients. PCR amplifies specific DNA sequences containing suspected mutations, producing millions of copies that can be analyzed. The amplified DNA can then be sequenced or examined for known cancer-associated mutations, making PCR essential in molecular diagnosis of genetic and cancer-related diseases.

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92. (3)

PCR stands for Polymerase Chain Reaction, *Taq* polymerase is isolated from *Thermus aquaticus*, and the process involves three steps—Denaturation (separation of DNA strands), Annealing (primer binding), and Extension (DNA synthesis). Statement B is false because PCR occurs *in vitro* (in test tubes), not *in vivo* (in living cells). Statement C is false because PCR uses primers (short DNA sequences) and DNA polymerase, not DNA ligase.

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93. (4)

27 documented varieties of Basmati rice are grown in India. Basmati rice is a premium, aromatic variety primarily cultivated in specific regions like Punjab and Haryana. These 27 varieties represent the official registered basmati cultivars recognized by agricultural authorities in India, distinguished by their unique aromatic characteristics and cooking properties.

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94. (3)

The gel was stained with ethidium bromide and exposed to UV radiation. Ethidium bromide is an intercalating dye that inserts between DNA base pairs. When exposed to UV radiation, the ethidium bromide fluoresces with a bright orange-red color, making DNA bands visible. This is the standard visualization method used in gel electrophoresis labs worldwide.

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95. (4)

List-I	List-II
Carcinogens	Agents which convert normal cells to cancerous cells
MRI	Useful to detect cancers of the internal organs
Radiotherapy	Tumor cells are irradiated lethally
α-interferon	Helps in destroying the tumor

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96. (4)

Denaturation involves the disruption of the secondary, tertiary, and quaternary structures of the protein. These higher-level structures are maintained by relatively weak bonds (hydrogen bonds, ionic bonds, and hydrophobic interactions) that are easily broken by heat or chemical changes. When these bonds break, the protein uncoils and loses its specific 3D shape, which is essential for its biological function.

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97. (4)

Biopiracy is the term used to refer to the use of bio-resources by multinational companies without proper authorization. This unethical practice involves taking advantage of traditional knowledge and biological resources from developing countries without compensation to local communities or nations. It violates biodiversity conventions and intellectual property rights, particularly affecting indigenous peoples and farmers in biodiverse regions.

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98. (3)

1. AIDS stands for Acquired Immune Deficiency Syndrome.
2. It is caused by a retrovirus called HIV (human immunodeficiency Virus).
3. HIV enters the body and affects macrophages that act as HIV- forming factories.
4. The newly formed virus destroys the helper-T cells and reduces the immunity of the individual.
5. There is always a time lag between the infection and the appearance of AIDS symptoms.
6. This asymptomatic period may vary from a few months to many years like 5-10 years.
7. Hence, both statements I and II are correct.
8. Hence, the correct answer is option (C).

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99. (4)

The integration of natural science and organisms for products and services as a definition of biotechnology was given by the European Federation of Biotechnology. This comprehensive definition encompasses all aspects of modern biotechnology, from genetic engineering to fermentation processes and pharmaceutical production, representing the broadest and most inclusive understanding of the field.

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100. (4)

Insulin cannot be administered orally to diabetic patients because it is a protein that would be digested in the gastrointestinal tract. Insulin is a 51-amino acid peptide hormone that would be broken down into smaller peptides and amino acids by proteolytic enzymes like pepsin and trypsin in the stomach and small intestine. This digestion would destroy its biological activity, making oral administration ineffective, necessitating injection instead.

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